

BALKHASH, KazakhstanWMO: **357960**Lat: **46.833N**Lon: **74.967E**Elev: **1148**StdP: **14.10**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-12.2	-23.4	1.6	-15.6	-19.2	2.0	-11.3	23.3	13.3	21.0	11.6	9.2	40	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	89.1	66.6	86.7	66.5	84.3	65.9	71.8	83.5	70.2	82.2	68.7	80.9	8.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	105.4	80.5	65.6	98.5	79.2	63.7	92.2	78.1	36.4	83.5	35.0	82.3	33.7	81.0	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	20.1	17.8	-21.9	95.8	4.9	3.7	-25.4	98.5	-28.3	100.6	-31.1	102.7	-34.7	105.3		
(5)			-21.7	75.4	5.1	2.2	-25.3	77.0	-28.3	78.3	-31.2	79.5	-34.9	81.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	7.3	11.2	28.5	47.8	60.8	71.6	74.2	71.5	59.7	45.7	29.7	14.9
	DBStd	25.19	10.89	11.12	11.72	8.83	6.76	5.97	5.12	5.83	7.38	7.96	10.23	10.98
	HDD50	5112	1325	1086	666	142	6	0	0	0	11	179	608	1089
	HDD65	8534	1790	1506	1130	517	161	14	3	14	188	599	1058	1554
	CDD50	2830	0	0	1	76	342	647	750	666	303	45	0	0
	CDD65	780	0	0	0	2	33	212	288	215	30	0	0	0
	CDH74	6923	0	0	0	16	210	1805	2756	1910	225	1	0	0
	CDH80	1891	0	0	0	3	28	467	831	513	49	0	0	0
Wind	WSAvg	9.1	9.3	9.5	9.9	9.3	9.6	9.4	9.0	9.1	8.8	8.6	8.5	8.7
Precipitation	PrecAvg	5.6	0.5	0.4	0.5	0.5	0.6	0.6	0.6	0.3	0.2	0.4	0.6	0.6
	PrecMax	9.8	1.3	1.0	1.2	1.3	2.3	3.5	3.7	1.8	3.0	1.4	1.6	1.5
	PrecMin	1.5	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.7	0.3	0.2	0.3	0.3	0.5	0.6	0.7	0.4	0.4	0.4	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	36.1	58.6	75.8	83.5	91.3	93.3	91.8	86.0	70.1	53.3	37.6
		MCWB	32.8	33.5	45.0	55.2	61.5	66.2	66.3	65.7	61.6	52.5	45.7	35.1
	2%	DB	30.7	33.6	52.3	68.9	79.1	87.6	89.5	87.9	79.6	65.5	49.4	34.3
		MCWB	30.0	32.0	43.4	52.1	60.3	66.3	67.6	66.1	60.2	51.7	44.2	32.7
	5%	DB	27.3	31.0	47.5	64.8	75.9	84.9	87.0	85.2	75.8	62.0	46.0	32.2
		MCWB	26.4	29.5	40.9	50.9	58.9	65.8	67.6	65.6	59.0	50.5	41.9	30.7
	10%	DB	23.3	28.0	43.2	61.4	73.1	82.4	84.5	82.7	72.6	58.3	42.9	29.6
		MCWB	22.4	26.6	38.5	49.5	58.1	65.5	66.9	64.8	57.7	48.7	39.4	27.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	34.5	48.2	58.5	66.0	72.7	74.9	73.3	67.4	57.3	49.0	35.5
		MCDB	33.4	35.3	54.8	69.8	77.4	83.9	86.5	85.0	78.2	64.4	51.6	36.5
	2%	WB	29.9	32.2	44.1	55.4	63.6	70.6	72.3	70.9	63.9	54.2	45.1	33.1
		MCDB	30.5	33.5	50.5	65.2	74.0	82.2	84.1	82.7	74.5	62.2	48.0	34.1
	5%	WB	26.4	29.6	41.3	52.9	61.7	68.9	70.8	68.9	61.6	51.8	42.3	30.9
		MCDB	27.1	30.8	46.9	62.0	72.3	80.7	82.9	81.2	72.6	59.7	45.5	32.1
	10%	WB	22.4	26.5	38.7	50.6	60.0	67.3	69.3	67.2	59.1	49.3	39.6	28.0
		MCDB	23.3	28.0	43.1	59.8	70.3	79.3	81.6	79.9	70.9	57.5	42.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.8	15.1	16.1	17.9	16.6	16.7	18.2	18.7	16.6	13.7	13.7
		MCDBR	13.3	14.7	18.6	21.1	21.2	19.2	19.4	20.7	22.0	20.7	16.3	12.7
		MCWBR	13.1	13.8	11.5	11.0	11.8	12.0	12.6	12.8	14.2	13.9	12.0	11.8
	5% WB	MCDBR	13.3	14.1	17.3	17.1	17.3	16.1	16.7	17.1	18.5	17.6	13.9	11.8
		MCWBR	13.1	13.7	11.3	10.3	11.8	11.9	12.6	13.7	15.3	13.7	11.4	11.4
Clear-Sky Solar Irradiance	taub	0.319	0.351	0.398	0.442	0.445	0.439	0.449	0.404	0.364	0.366	0.346	0.317	
	taud	2.026	1.968	2.048	2.034	2.081	2.167	2.167	2.156	2.270	2.336	2.233	2.156	2.073
	Ebn at Noon	229	247	256	257	261	263	258	266	268	244	220	214	
	Edh at Noon	32	42	45	50	50	46	46	39	34	32	28	27	
All-Sky Solar Radiation	RadAvg	451	733	1151	1765	2187	2285	2182	2024	1583	970	529	334	
	RadStd	83	69	122	101	100	122	133	78	72	73	52	62	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page